

PRE-CLASS READING 4 OF 4

Two ways AI enters an industry — replace the computation, or create the measurement

GraphCast — Lam, R. et al., “Learning skillful medium-range global weather forecasting,” *Science*, 14 November 2023 (read the Google DeepMind blog post), with its successors **GenCast** (*Nature*, December 2024) and **WeatherNext 2** (17 November 2025). **Climate TRACE** — the emissions-monitoring coalition founded in 2020, now 100+ contributing organisations; read the “about the data” note and one monthly release. Two flagship “AI for the planet” projects that look similar and create value in completely opposite ways — which is why the OBTL+ outline pairs them, and why this brief treats them as one argument.

ASSIGNED READING

Read the **GraphCast blog post** from Google DeepMind (≈ 8 min) and the **WeatherNext 2 announcement** (≈ 5 min) — the *Science* paper is optional and technical. Then read **Climate TRACE’s current data note** and skim **one monthly release** (≈ 8 min). Do not read them as two environmental stories. Read them asking a single structural question: **in each case, what existed before, and what did AI replace or create?** The answers are different, and the difference is the most portable thing in this week’s reading.

» sources ~21 min • this summary + reasoning notes ~23 min

Note on currency. GraphCast is the 2023 landmark and it has already been superseded twice by its own authors — **GenCast** (December 2024) added probabilistic ensembles, and **WeatherNext 2** (17 November 2025) added a Functional Generative Network that produces hundreds of scenarios in under a minute on a single chip, beating the previous model on **99.9% of variables and lead times** and running **8× faster**. Read GraphCast for the mechanism and the argument; do not quote its 2023 accuracy figures as current. Climate TRACE moves faster still: it now publishes **monthly**, roughly two months in arrears, so any figure you cite needs its release version attached.

KEY TAKEAWAY

These two projects sit in the same sentence in every “AI and climate” article and they are not the same kind of thing at all. **GraphCast replaced a computation that already existed.** Numerical weather prediction has been solving physics equations on supercomputers for decades; GraphCast learned to approximate the answer directly, producing **more accurate predictions on more than 90% of 1,380 tested variables and lead times** than the European Centre’s HRES system, and making a **10-day forecast in under a minute on a single TPU** where the conventional method takes hours on hundreds of machines. Nothing new is being measured — a forecast at least as good arrives for a tiny fraction of the compute (DeepMind publishes the two run times, not a ratio; the implied saving is roughly three orders of magnitude). **Climate TRACE created a measurement that did not exist.** Greenhouse-gas emissions were self-reported, annual, and often years late; there was no independent meter. Climate TRACE infers emissions from observable proxies — satellite imagery, thermal signatures, ship transponders, plant-level activity data — and now publishes monthly estimates covering **2,765,771 emissions sources summarised from about 745 million assets across nine sectors**. Nothing got faster; something became **visible**. Those are the two archetypes, and almost every AI use case in any vertical is one or the other: **you are either replacing an expensive computation you already trusted, or manufacturing a measurement that nobody had.** They have different preconditions, different risks, and completely different ways of failing.

WHY IT MATTERS FOR THIS COURSE

Capsule 3 asks you to pattern-match AI opportunities into your own industry — healthcare, finance, manufacturing, mobility, environment — and the failure mode of that exercise is producing a list of applications rather than a way of thinking. These two archetypes are the way of thinking. They also settle the **data-readiness** question that Capsule 1 opens with, in the most concrete way available: GraphCast was possible because **ERA5** existed — four decades of standardised, quality-controlled, freely available global reanalysis in one consistent format. That dataset is the entire precondition, and it took a public institution decades to build. When you ask “is our data ready?”, the honest version of the question is “**what is our industry’s ERA5, and if there isn’t one, who would have to build it?**” Most industries do not have one, and finding that out is a finding, not a failure.

Feeds Class 5: C1 (data readiness • what a use case requires before it is even possible) • C3 (vertical applications • pattern-matching to your own industry • the case discussion).

MAIN POINTS

Archetype 1 • GraphCast — the learned surrogate

- **What it is.** A graph neural network that takes the current state of the atmosphere and the state six hours ago, and predicts the state six hours ahead — then feeds its own output back in, autoregressively, to reach ten days.

0.25° resolution (about 28 km at the equator, over a million grid points), **five surface variables and six atmospheric variables across 37 altitude levels**.

- **The result.** More accurate than the European Centre's HRES on **more than 90% of 1,380 test variables and lead times**, and on **99.7%** of tested variables in the troposphere — the layer that matters for weather people experience. In one widely-cited instance it placed **Hurricane Lee's landfall in Nova Scotia about nine days ahead**, against roughly six days for conventional forecasting.
- **The economics, which are the actual story.** A ten-day forecast in **under a minute on a single Google TPU v4**, against hours on a supercomputer running hundreds of machines. DeepMind does not publish a cost ratio; a rough reading of those two statements is three orders of magnitude, and it should be treated as an estimate. DeepMind **open-sourced the code**, with the European Centre for Medium-Range Weather Forecasts beginning to run its forecasts experimentally.
- **The precondition — read this twice.** GraphCast was trained on about **four decades of ERA5**, the ECMWF's reanalysis dataset: a physically consistent, gridded, quality-controlled reconstruction of global weather, standardised, public and free. **No ERA5, no GraphCast.** The model is a few months of research; the dataset is a multi-decade public investment by an intergovernmental institution.
- **And the uncomfortable part.** ERA5 is itself the *output of a physics model* assimilating observations. So GraphCast is a surrogate trained on a simulation — it learns the simulator's biases along with its skill, and it cannot in principle be more right than the system it learned from in regimes where that system is wrong. Which is exactly where a changing climate takes you.

What the successors fixed, and why it is a design lesson

- **The problem with GraphCast.** A model trained to minimise average error produces a *blurry* forecast: hedging toward the mean is the optimal strategy for that objective, so extremes get smoothed. For weather, the extremes are the entire point. **The metric you optimise is the behaviour you get** — the same lesson as Week 3's eval design, in a different industry.
- **GenCast (December 2024)** moved to a **diffusion-based ensemble**: instead of one best-guess trajectory, generate many plausible ones and read the probability off the spread. This is what forecasters actually need — not "it will rain", but "a 30% chance, and here is the tail".
- **WeatherNext 2 (17 November 2025)** introduced a **Functional Generative Network**, injecting noise into the model architecture itself so that each sampled scenario stays *physically coherent across variables* rather than being independently noisy per variable. Result: **hundreds of scenarios in under a minute on a single TPU**, hourly resolution out to **15 days, 8× faster** than its predecessor and better on **99.9% of variables and lead times**.
- **Where it ended up.** Not as a research artefact — inside **Google Search, Gemini, Pixel Weather and the Maps Platform Weather API** (with Google Maps itself announced as coming), and available through Earth Engine, BigQuery and Vertex AI. This is the vertical-application arc in miniature: research result → open weights → cloud product → consumer feature, in roughly two years.

Archetype 2 · Climate TRACE — the manufactured measurement

- **The problem it attacks.** Emissions were **self-reported** by the entity emitting them, aggregated nationally, published annually, often two or more years late, and effectively unverifiable. Every climate commitment in the world rested on numbers that nobody independently checked. Note the shape: this is not a slow computation, it is a **missing meter**.
- **The method.** Infer emissions from things that can be *observed from outside*: satellite imagery, thermal and spectral signatures, radar, ship and aircraft transponder tracks, plant-level activity data — then calibrate those proxies against the cases where real measurements exist, and extrapolate. AI is the inference layer that turns an observable proxy into an emissions estimate at scale.
- **The scale it reaches.** The May 2026 release (v5.9.0, published 30 July 2026) covers **2,765,771 emissions sources summarised from 744,678,997 assets**, across **nine sectors** — agriculture, buildings, fluorinated gases, fossil-fuel operations, manufacturing, mineral extraction, power, transportation and waste — including the **100 largest urban areas**, with **monthly data roughly two months in arrears**. The coalition includes Carnegie Mellon, Duke, Global Energy Monitor, RMI and WattTime among **more than 100 contributing organisations**.
- **What that release actually said**, so you can see the granularity: global emissions of **5.0 billion tonnes CO₂e for the month, down 0.1% year-on-year**; China at **1.4 billion tonnes, down 0.6%**; methane at **34.2 million tonnes, up 0.1%**; buildings the largest sectoral increase at **+1.6%**. Monthly, per-country, per-sector, per-facility — where the alternative was an annual national total published in arrears.
- **The precondition here is different.** Not a historical dataset, but an **observable proxy plus a calibration set**. Emissions work because burning things is visible from orbit. The question to ask of any use case in this archetype is: *is the thing I want to measure connected to something an instrument can see, and do I have enough ground truth to calibrate the link?*

The comparison, which is the point of pairing them

- **What existed before.** GraphCast: a trusted, expensive computation. Climate TRACE: a self-reported number nobody could check.
- **What AI supplied.** GraphCast: the *same answer*, ~1,000× cheaper and faster. Climate TRACE: *an answer at all*, independent of the party being measured.
- **What the value is.** GraphCast: speed and cost, which convert into ensembles, more frequent runs, and forecasting on a laptop instead of a national facility. Climate TRACE: **accountability**. The value is not that the number is better; it is that it is produced by someone other than the emitter.
- **How each fails.** The surrogate fails by inheriting its teacher's blind spots and by smoothing exactly the events you cared about. The manufactured measurement fails by being an **estimate presented as a fact** — a proxy with uncertainty bounds that get dropped the moment the number reaches a headline.
- **What each needs before you start.** Surrogate: a long, standardised, labelled history — and an existing expensive computation worth replacing. Manufactured measurement: an observable proxy, a calibration set, and an institutional willingness to accept a number that the measured party did not produce.

Third archetype, from Week 1: the capability that had no prior at all — and it is worth unpacking, because it is the one everybody wants to claim.

AlphaFold is neither of the two above, and the reason is exactly the phrase “no computation and no measurement”. *No computation*: predicting a protein's three-dimensional shape from its amino-acid sequence was a **fifty-year-old open problem** — not an expensive method waiting to be made cheap, but no reliable method at all. Nothing to build a surrogate of. *No practical measurement*: determining a structure experimentally works, but far too slowly to cover the field — **about 200,000 structures since 1958, against more than 200 million known proteins**. AlphaFold DB opened at ~350,000 predictions and passed **200 million within about a year**: sixty-five years of laboratory work against one of inference. Unlike Climate TRACE, which had an observable proxy to infer from, there was **no route to the answer at any price** — which is what makes it a third category rather than a harder archetype two.

Two consequences, and the second is the one that will decide whether your proposal survives a finance review.

First, the diagnostic: **name what existed before**. A trusted expensive computation → archetype one. An unverifiable self-report, or no measurement at all → archetype two. Neither → this third category, rare, or far more likely **you have not yet defined the problem**. Second, and this is where Reading 3 comes back: **A and B have a comparator and this one does not**. GraphCast can be priced against the supercomputer run it replaces; Climate TRACE against the audit regime that never existed. Here **nothing was being spent, so there is no saving to compute**. An archetype-three case can only ever be argued as *value creation*, never as *cost to serve* — which is the hardest argument to put in front of a CFO, and the reason to be honest about which of the three you are actually in.

WHERE TO BE SCEPTICAL

Both projects are genuinely impressive and both are routinely over-claimed. Four things worth holding onto when someone puts either on a slide.

1. **A surrogate cannot exceed its teacher where the teacher is wrong.** GraphCast learned from ERA5, which is itself a physics model assimilating observations. In atmospheric regimes outside the training distribution — which is precisely what a changing climate produces — there is no reason to expect the learned model to extrapolate correctly, and good reason to expect it not to. Speed is not the same as understanding.
2. **“Beats HRES on 90% of variables” is a benchmark claim.** It is measured against reanalysis, on selected variables and lead times, under a fixed evaluation protocol. It is a strong result and it is the same species of claim as the SWE-bench number in Week 4: a lower bound on capability that says nothing about whether it is useful for the decision you need to make. Operational meteorologists still run both.
3. **Climate TRACE produces estimates, not measurements.** Its numbers frequently differ from official national inventories, and it does not have the authority to settle that disagreement — it has the *evidence* to start it. Uncertainty ranges are published and almost never survive into the reporting. A per-facility monthly estimate looks like a meter reading and is not one.
4. **Neither project is a business.** GraphCast's value accrues to Google as a product feature and to the public as an open model; Climate TRACE is a philanthropically funded coalition. When you pattern-match these into your own industry, you are borrowing the *shape* of the value, not a proven commercial model — and Reading 3 is a reminder that the shape and the margin are separate questions.

READING FOCUS QUESTIONS — bring your answers to class

There are no wrong answers here — these surface how you think, not what you recall. A clue sits under each question to give you a direction to push on; the fuller reasoning is in the section after.

- Both projects are filed under “AI for climate” and they create value in opposite ways. State the difference in one sentence — then classify the best AI use case in your own industry as one archetype or the other.

Clue. Ask one question of each project and let it sort them: **what existed before?** If the answer is “an expensive computation we already trusted”, you are looking at a surrogate; if it is “a number somebody self-reported and nobody checked”, you are looking at a manufactured measurement. Then apply the same question to your own candidate use case, and be suspicious if you want to answer “neither, ours is genuinely new” — that third archetype is real and it is rare.

- GraphCast exists because ERA5 exists. What is your industry’s ERA5 — and if there is not one, what does that tell you about your AI roadmap?

Clue. Be strict about what qualifies. Not “we have a lot of data” — a dataset of this kind is **long, standardised, quality-controlled, consistently formatted, and accessible to the people who would build on it**. Test your candidate against all five. If it fails on standardisation or access rather than on volume, notice that those are governance problems rather than technical ones, and ask who would have to decide to fix them.

- A learned surrogate is trained on the output of the simulator it replaces. What can it never do that the simulator can — and in what circumstances does that matter enough to keep the old system?

Clue. Start from what the physics model actually contains that the network does not: conservation laws that hold whether or not similar conditions appeared in the training data. Then ask what happens to each system when conditions arrive that are outside anything in forty years of history. Connect this to Week 4’s distinction between predicting the next token and predicting a world state — it is the same argument in a different field.

- Climate TRACE’s figures often disagree with a country’s own reported inventory. Who wins that argument — and what would it take for an independently manufactured measurement to become authoritative?

Clue. Notice that this is not a question about accuracy. Trace what makes any measurement authoritative — who produces it, whether the method is open, whether it can be independently reproduced, whether an institution has agreed to be bound by it — and check Climate TRACE against each. Then look for the historical parallels, because credit ratings, financial audit and nutrition labelling all had to travel this road, and none of them travelled it by being right.

- A ten-day forecast went from hours on a supercomputer to under a minute on one chip. Name something that becomes possible at that price which is not simply “the same thing, faster”.

Clue. A thousandfold cost reduction usually changes the kind of question you can ask, not just the speed of the old one. Ask what you would do if a forecast were nearly free: how many would you run, and why would you run more than one? WeatherNext 2 has already answered — look at what it produces in that minute and ask what it is for. Then generalise: when a computation gets a thousand times cheaper in your own industry, what would you compute that nobody computes today?

HOW TO THINK ABOUT THESE — read after you have had a go

These are not model answers to be memorised — they are the line of reasoning we will pick up in class. Class time is short, so they are written to stand on their own.

Q1 — Cheaper answer versus first answer

One sentence: **GraphCast makes an answer we already had dramatically cheaper — minutes on one chip against hours on a supercomputer; Climate TRACE produces an answer that did not previously exist.** Everything else follows from that. The surrogate archetype needs an incumbent computation worth replacing and a long labelled history of its outputs, and its risk is inheriting the incumbent’s errors. The manufactured-measurement archetype needs an observable proxy and a calibration set, and its risk is that an estimate gets treated as a fact. When you classify your own use case, the “what existed before?” question is deliberately blunt because it stops the usual drift into “AI will transform our industry”. If your industry runs an expensive simulation, optimisation or model — pricing, risk, logistics routing, structural analysis, chip verification, clinical trial simulation — you are almost certainly in archetype one, and your first question is whether you have decades of its inputs and outputs stored in a usable form. Most organisations have run the computation for years and thrown the results away, which is a painful realisation and a cheap fix going forward. If your industry runs on numbers that the interested party supplies — supplier compliance, insurance risk declarations, self-reported ESG, loan applications, warranty claims — you are in archetype two, and your first question is whether anything observable from outside correlates with the thing being declared. **Archetype two has a second and commoner flavour: not a number somebody shaded, but one that never existed for most cases at all**, because measuring it took a scarce person or a slow instrument and was only ever done for a sample. Three examples. **Healthcare:** grading a retinal photograph for diabetic eye disease needs an ophthalmologist, so in most health systems the majority of diabetic patients never had the result at all — a model reading the photograph does not make a diagnosis cheaper, it creates one where there was none. **Telecom:** operators have always measured the *network* — throughput, drop rates, cell KPIs — while what a given subscriber experienced was known only from complaints, a self-selected sample of the angriest. Inferred from signalling and handset telemetry, it becomes a number for every customer. **Manufacturing:** defect inspection was a sample, or a

person checking everything badly; per-unit vision inspection creates a record for every unit where one existed for one in N. In all three the proxy was **already being collected** — nobody had turned it into a measurement. Notice, and this is the link to Capsule 3, that the case we discuss is squarely archetype two where it matters: **a credit score built from observed payment behaviour, for a population that had no credit history at all.** (Its payments business is archetype one — a settlement computation banks already ran, made radically cheaper — so the same company ran both, which is worth noticing when you screen your own portfolio.) Ant did not make lending decisions faster. It manufactured a measurement of creditworthiness where the incumbent system had none, for the **729 million** Alipay users who took up at least one of its financial products — and the value, the moat and the eventual regulatory problem all came from that single fact.

Q2 — Long, standardised, clean, consistent, accessible — and most industries fail on the last two

ERA5 passes all five tests: roughly forty years, a single consistent grid and format, quality-controlled by an intergovernmental institution, physically coherent, and free to download. That combination is extraordinarily rare and it is worth being blunt about why. It exists because weather forecasting is a public good funded by governments that agreed on standards, not because meteorologists are better at data engineering. Run the five tests honestly on your own industry and the usual result is a pass on volume and a failure on standardisation and access — the data exists, in eleven systems, in six formats, with three incompatible definitions of the key entity, owned by four departments who have never had to reconcile them. That is a **governance** finding, not a technical one, and it is the most useful thing a data-readiness assessment can produce, because it tells you the first project is not a model at all. There are real examples of industries that built their ERA5 and were transformed by it: genomics has public reference genomes and sequence archives; radiology has curated labelled image sets; finance has decades of standardised market ticks. And there are industries that did not, and whose AI stories are correspondingly thin — construction, most of manufacturing maintenance, and large parts of clinical practice where the record is unstructured narrative. So the roadmap implication is uncomfortable and correct: **if your industry has no ERA5, then either your first AI project is a data-standardisation project, or your realistic ambition is confined to use cases that run on somebody else's data — which means on capability your competitors can rent too.** That is not defeatism. It is the difference between a three-year plan and a three-year disappointment, and knowing which one you are in is worth a great deal.

Q3 — It cannot obey a law it was only ever shown examples of

The physics model contains conservation of mass, momentum and energy as *constraints*, holding whether or not anything resembling the current situation has occurred before. The neural surrogate contains a statistical regularity extracted from forty years of examples. Inside the distribution those look identical and the surrogate is often better, because it has also absorbed the systematic biases of the observation network in a way that helps it fit reality. Outside the distribution they part company completely: the simulator degrades gracefully because the laws still apply, while the surrogate has no notion that it has left familiar territory and will produce a confident, physically impossible field with no signal that anything is wrong. This is the same distinction Week 4 drew between a system predicting the next token and a system predicting a world state, and it is worth noticing that a research community with every incentive to declare victory has instead kept the physics models running. When does it matter enough to keep the old system? Three conditions, and they generalise well beyond weather. **When the tail is the point** — you are forecasting precisely to catch the unprecedented event, which is by construction outside the training data. **When you need the intermediate quantities** — the simulator tells you *why*, the surrogate tells you *what*, and in a regulated or safety-critical setting the *why* is often the deliverable. **When the underlying system is changing**, so yesterday's statistics are a decaying guide, which is the specific and slightly bleak irony of using climate-era weather data to train a model whose main application is a changing climate. The practical resolution in operational meteorology is the sensible one and it transfers: **run both, use the cheap surrogate for volume and the expensive simulator as the check**, and treat disagreement between them as a signal worth investigating rather than an inconvenience. If you take one design pattern from this reading into your project, take that one.

Q4 — Authority is institutional, not empirical

The trap in this question is assuming that the more accurate number wins. It does not, and the history is unambiguous about it. What makes a measurement authoritative is a chain of institutional properties: an **open, documented method** that a third party can reproduce; **independence** from the party being measured; **consistency over time**, so that a change in the number means a change in the world rather than a change in the method; **acceptance by an institution with consequences attached** — a regulator, a standard-setter, an exchange, a court. Climate TRACE currently has the first two comprehensively, publishes its methodology and its uncertainty, and is improving on the third as its version history lengthens. What it does not have is the fourth. Nobody is obliged to act on its numbers, which is why disagreement with a national inventory does not resolve — the two claims sit side by side, and the official one retains standing. The historical parallels are worth carrying into class because they show the shape of the road. Financial audit became authoritative when securities regulators required it, not when accountants got better. Credit ratings acquired force when capital rules referenced them — and, instructively, the 2008 crisis showed what happens when an authoritative measurement is paid for by the party it measures, which is exactly the independence property Climate TRACE has and traditional inventories lack. Nutrition labelling required legislation. In each case the sequence was the same: an independent measurement becomes available and is

disputed; it becomes cheap and continuous enough to be routine; some party with an interest in accuracy — an insurer, a lender, a buyer imposing supply-chain requirements — starts using it in contracts; and only then does an institution codify it. **The real question for your own industry is not whether an AI-manufactured measurement can be accurate. It is who would have a commercial reason to be bound by it, and that is usually a party other than the one being measured.** In the Capsule-3 case, that party turned out to be the state, and the resolution was to license the measurement and take a stake in it.

Q5 — At a thousandth of the price you stop asking for one forecast

The direct answer is **ensembles**, and the whole subsequent history of the field is the field realising it. If a forecast costs hours of supercomputer time, you run one, or a handful, and you present a single trajectory because that is what you can afford. If it costs a minute on one chip, you run hundreds — and hundreds of forecasts is not the same output faster, it is a **different product**: a probability distribution instead of a point estimate. That is what a decision-maker actually needs. “Rain on Thursday” cannot be acted on; “a 30% chance of rain on Thursday, with a 5% tail of something severe” can, because it can be weighed against the cost of preparing. WeatherNext 2 produces exactly this — hundreds of physically coherent scenarios in under a minute — and it is no accident that its architecture is generative: once you can afford many samples, the right model is one that samples well rather than one that averages well. Two further things unlock at that price and both generalise. **Frequency**: cheap forecasts can be re-run continuously as new observations arrive, so the forecast becomes a live signal rather than a daily publication. **Access**: at a minute per run on commodity hardware with open weights, forecasting stops being the exclusive capability of nations with supercomputers — which is a substantial redistribution of a public good toward the countries most exposed to weather risk and least able to fund a national centre. Before the business version, three from ordinary life, because everyone has lived them. **Photography**: when a frame cost money and you waited a week to see it, you took twenty-four pictures on a holiday and composed every one. When the marginal cost reached zero people did not simply take more holiday photographs — they began photographing receipts, parking-bay numbers, whiteboards, wine labels. The camera became a **memory tool**, which is a different device. **The international call**: when it cost real money by the minute you planned it, kept it short, and it was an event; at effectively nothing you ring to say nothing in particular, and the call becomes ambient contact rather than a transaction. **Storage**: when a terabyte was expensive you deleted the logs after thirty days; when it was cheap you kept everything — and keeping everything is the precondition for every machine-learning project your organisation now wants to run. In each case the cheap thing was not used *more*. It was used **differently**. Now generalise the pattern, because this is the transferable question. **When a computation in your industry gets a thousand times cheaper, the win is rarely doing the old thing faster — it is doing something you never did at all because doing it once was pointless.** Risk models run per-customer instead of per-portfolio. Simulations run across a thousand design variants instead of the three someone had time to specify. Every document checked rather than a 2% sample. Each of those is a change in the unit of analysis, and the unit of analysis is where the value is. Two economic ideas are at work here and they are worth keeping apart. **Jevons’ paradox** — the one Reading 3 leans on — says that when a resource gets cheaper, total consumption rises far enough that the bill does not fall: a claim about **volume and cost**. This question is about something else — that a large enough price fall changes the **class of question you can afford to ask**: a claim about **kind**. They are related and they are not the same. **Jevons explains why your invoice went up. This explains why that can be the right outcome** — you did not buy more of the old thing, you bought a question you could not previously ask.

APPENDIX — VOCABULARY, EXPLAINED

*The distinction that carries this reading: replacing a computation you already trusted, and creating a measurement that did not exist. It does two jobs, and the second is the more useful one. With a candidate already in hand it is a **classification** — which of the two is this, and therefore what must I have before I start? With no candidate yet it is **two separate searches of the same organisation**: walk it once asking where is there an expensive computation we already trust?, then walk it again asking where is there a number nobody credibly measures? You are not choosing between them — a sensible portfolio holds both, and they need completely different things to succeed. Every term below belongs to one or the other.*

Numerical weather prediction (NWP) and HRES

- **Plain.** Solving the physics of the atmosphere on a supercomputer to work out what happens next.
Technically. Discretise the governing equations on a global grid and integrate forward in time. **HRES** is the European Centre’s high-resolution deterministic forecast, the industry benchmark GraphCast was measured against. Cost: hours on hundreds of machines per run.

Reanalysis, and ERA5

- **Plain.** A best reconstruction of what the weather actually was, everywhere, for the last forty years — built by fitting a physics model to every observation ever recorded.
Technically. ECMWF’s **ERA5**: consistent, gridded, quality-controlled global reanalysis, freely available. It is GraphCast’s training data and therefore its ceiling — a surrogate trained on a reanalysis inherits the assimilating model’s biases. **The single most important precondition in this reading.**

Surrogate model (emulator)

- **Plain.** A fast approximation that learned to imitate a slow, expensive calculation.
Technically. Trained on input-output pairs from the expensive system. Gains orders of magnitude in speed; loses the guarantees the original system encoded, and generalises poorly outside the training distribution. The dominant pattern for AI in scientific and engineering verticals.

Deterministic vs ensemble forecast

- **Plain.** One answer, versus many answers and how often each occurred.
Technically. A deterministic model trained on mean squared error hedges toward the average, which **smooths exactly the extremes you needed**. An ensemble samples many trajectories and reads probability off the spread. GenCast (Dec 2024) and WeatherNext 2 (Nov 2025) are the field's move from the first to the second.

Functional Generative Network (FGN)

- **Plain.** Randomness built into the model itself, so each scenario it produces hangs together physically.
Technically. WeatherNext 2's architecture: noise injected into the network rather than added to outputs, so sampled scenarios remain coherent *across variables* rather than independently noisy per variable. Enables hundreds of scenarios in under a minute on one TPU.

Lead time and skill

- **Plain.** How far ahead you are predicting, and how much better than a naive guess you are.
Technically. Forecast quality is always reported as a pair. GraphCast's "90% of 1,380 variables" is a count over the **variable × lead-time grid** — which is why the headline needs both numbers to mean anything, and why per-variable results matter more than the aggregate.

Ground truth vs proxy measurement

- **Plain.** Measuring the thing, versus measuring something visible that reliably moves with it.
Technically. Climate TRACE cannot meter a smokestack from orbit; it observes thermal, spectral and activity signatures and infers emissions via models calibrated on the minority of cases where direct measurement exists. Every proxy measurement carries calibration error and extrapolation risk, and both are usually dropped by the time the number is quoted.

MRV — measurement, reporting, verification

- **Plain.** Who measures it, who says it, and who checks.
Technically. The climate-policy term for the accountability chain. The traditional chain has the emitter performing all three roles. Climate TRACE's contribution is to separate measurement from the measured party — an **independence** property, which is a different kind of improvement from an accuracy one.

Bottom-up vs top-down inventory

- **Plain.** Add up what each facility says it emitted, or observe the whole system from outside and work backwards.
Technically. National inventories are bottom-up and self-reported. Climate TRACE is asset-level observation aggregated upward — nominally bottom-up, but built from external observation rather than declaration, which is what makes it independent. Disagreement between the two is expected and is the point.

The observability precondition

- **Plain.** You can only manufacture a measurement of something that leaves a trace an instrument can see.
Technically. The screening test for archetype two: is the quantity of interest causally linked to an observable signal, and do you have enough labelled cases to calibrate the link? Emissions pass because combustion is visible from orbit. Many attractive-sounding use cases — culture, intent, effort — fail here and no model fixes it.

SOURCE LINKS

Companion readings in this pack — these are the "Reading N" references used throughout this brief. **Reading 1** — Iansiti & Lakhani, *Competing in the Age of AI* (the operating-model framework and the AI factory). **Reading 2** — McKinsey, *The State of AI* (who was asked, what threshold, what denominator). **Reading 3** — *AI unit economics and cost to serve* (the arithmetic behind any cost claim). **Case pre-read 1** — *Ant Group (A)*, the Capsule-3 case and this brief's archetypes in the wild: **B** on credit, **A** on payments.

Google DeepMind — "GraphCast: AI model for faster and more accurate global weather forecasting" (Nov 2023) • deepmind.google/blog/graphcast-ai-model-for-faster-and-more-accurate-global-weather-forecasting/ (the assigned reading)

Lam et al. — "Learning skillful medium-range global weather forecasting", *Science* (14 Nov 2023) • www.science.org/doi/10.1126/science.adj2336 (optional — the peer-reviewed paper)

Google — "WeatherNext 2: our most advanced forecasting model" (17 Nov 2025) • blog.google/innovation-and-ai/models-and-research/google-deepmind/weathernext-2/ (the current state of the art)

Climate TRACE — data downloads and methodology • climatetrace.org/data

Climate TRACE — May 2026 emissions data release (published 30 Jul 2026) · climatetrace.org/news/climate-trace-releases-may-2026-emissions-data *(the figures quoted here)*

Companion · **Reading 1**'s data-pipeline component is what ERA5 is an extreme example of; **Reading 3** is why an order-of-magnitude fall in the cost of a computation changes what you compute rather than what you spend.

CAVEATS

1. This summary is not a substitute for the sources, and the DeepMind blog posts in particular are short and worth reading in full — the diagrams do work the prose cannot.
2. GraphCast's 2023 figures are quoted as published and are **no longer the state of the art**; its own successors have superseded them twice. Cite them as a landmark, not as current performance.
3. Climate TRACE figures are **estimates with published uncertainty**, versioned monthly. Any number quoted from it needs its release version and date attached; the figures here are from v5.9.0, covering May 2026, published 30 July 2026.
4. Neither project is a commercial model you can copy. The archetypes transfer; the funding, the institutional position and the margin structure do not.
5. Generated and fact-checked using Anthropic models. The archetype framing, the data-readiness screen and the "authority is institutional" argument are the instructor's reading, not claims made by either project. Flag anything that looks wrong.